№2

1) $\int \frac{x}{x^2+x+1}\,dx = \frac{1}{2}\int \frac{2x+1}{x^2+x+1}\,dx - \frac{1}{2}\int \frac{1}{x^2+x+1}\,dx = \frac{1}{2}\int \frac{d(x^2+x+1)}{x^2+x+1} - \frac{1}{2}\int \frac{1}{x^2+x+1}\,dx =$

$= \frac{1}{2}\ln|x^2+x+1| - \frac{1}{2}\int \frac{1}{x^2+x+1}\,dx$ (=)

$\int \frac{1}{x^2+x+1}\,dx = \int \frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}}\,dx = \left\{ \begin{array}{l} t = x+\frac{1}{2} \\ dt = dx \end{array} \right\} = \int \frac{dt}{t^2+\frac{3}{4}} = \frac{2}{\sqrt{3}}\operatorname{arctg}\frac{2}{\sqrt{3}}t + C =$

$= \frac{2}{\sqrt{3}}\operatorname{arctg}\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right) + C$

(=) $\frac{1}{2}\ln|x^2+x+1| - \frac{1}{\sqrt{3}}\operatorname{arctg}\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right) + C = \frac{\ln(x^2+x+1)}{2} - \frac{\operatorname{arctg}\frac{2x+1}{\sqrt{3}}}{\sqrt{3}} + C$

2) $\int \frac{x}{\sqrt{5x^2-2x+1}}\,dx = \int \frac{x}{\sqrt{(\sqrt{5}x-\frac{1}{\sqrt{5}})^2+\frac{4}{5}}}\,dx = \left\{ \begin{array}{l} t = \sqrt{5}x - \frac{1}{\sqrt{5}} \\ x = \frac{t+\frac{1}{\sqrt{5}}}{\sqrt{5}} = \frac{\sqrt{5}t+1}{5} \\ dx = \frac{1}{\sqrt{5}}dt \end{array} \right\} =$

$= \int \frac{\frac{\sqrt{5}t+1}{5}}{\sqrt{t^2+\frac{4}{5}}}\cdot\frac{1}{\sqrt{5}}\,dt = \int \frac{t+\frac{1}{\sqrt{5}}}{5\sqrt{t^2+\frac{4}{5}}}\,dt = \frac{1}{5}\int \frac{t}{\sqrt{t^2+\frac{4}{5}}}\,dt + \frac{1}{5\sqrt{5}}\int \frac{dt}{\sqrt{t^2+\frac{4}{5}}} =$

$= \frac{1}{5}\left(\int \frac{t}{\sqrt{t^2+\frac{4}{5}}}\,dt + \frac{1}{\sqrt{5}}\ln\left|t+\sqrt{t^2+\frac{4}{5}}\right|\right) = \frac{1}{5}\left(\frac{1}{2}\int \frac{d(t^2+\frac{4}{5})}{\sqrt{t^2+\frac{4}{5}}} + \frac{1}{\sqrt{5}}\ln\left|t+\sqrt{t^2+\frac{4}{5}}\right|\right) =$

$= \frac{1}{5}\left(\sqrt{t^2+\frac{4}{5}} + \frac{1}{\sqrt{5}}\ln\left|t+\sqrt{t^2+\frac{4}{5}}\right|\right) + C =$

$= \frac{1}{5}\left(\sqrt{5x-2x+1} + \frac{1}{\sqrt{5}}\ln\left|\sqrt{5}x - \frac{1}{\sqrt{5}} + \sqrt{5x-2x+1}\right|\right) + C$

3) $\int \frac{3x^2+2x-3}{x^3-x}\,dx = \int \frac{3x^2+2x-3}{x(x-1)(x+1)}\,dx =$

$$\frac{3x^2+2x-3}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+1} = \frac{A(x-1)(x+1)+B(x+1)x+C(x-1)x}{x(x-1)(x+1)} =$$

$$= \frac{A(x^2-1)+B(x^2+x)+C(x^2-x)}{x(x-1)(x+1)}$$

$$\begin{cases} A+B+C=3 \\ B-C=2 \\ -A=-3 \end{cases} \qquad \begin{cases} A=3 \\ B=1 \\ C=-1 \end{cases}$$

$$= 3\int\frac{dx}{x} + \int\frac{dx}{x-1} - \int\frac{dx}{x+1} = 3\ln|x| + \ln|x-1| - \ln|x+1| + C$$

4) $\int\frac{dx}{x^4+1} = \int\left(\frac{x^2+1}{2(x^4+1)} - \frac{x^2-1}{2(x^4+1)}\right)dx = \frac{1}{2}\int\frac{x^2+1}{x^4+1}\,dx - \frac{1}{2}\int\frac{x^2-1}{x^4+1}\,dx =$

$$\int\frac{x^2+1}{x^4+1}\,dx = \int\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\,dx = \int\frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}\,dx = \left\{\begin{array}{l} t = x-\frac{1}{x} \\ dt = \left(1+\frac{1}{x^2}\right)dx \end{array}\right\} =$$

$$= \int\frac{dt}{t^2+2} = \frac{1}{\sqrt{2}}\operatorname{arctg}\frac{t}{\sqrt{2}} + C = \frac{1}{\sqrt{2}}\operatorname{arctg}\frac{x-\frac{1}{x}}{\sqrt{2}} + C$$

$$\int\frac{x^2-1}{x^4+1}\,dx = \int\frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\,dx = \int\frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)^2-2}\,dx = \left\{\begin{array}{l} t = x+\frac{1}{x} \\ dt = \left(1-\frac{1}{x^2}\right)dx \end{array}\right\} =$$

$$= \int\frac{dt}{t^2-2} = -\int\frac{dt}{2-t^2} = -\frac{1}{2\sqrt{2}}\ln\left|\frac{\sqrt{2}+t}{\sqrt{2}-t}\right| + C = -\frac{1}{2\sqrt{2}}\ln\left|\frac{\sqrt{2}+x+\frac{1}{x}}{\sqrt{2}-x-\frac{1}{x}}\right| + C$$

$$= \frac{1}{2}\cdot\frac{1}{\sqrt{2}}\operatorname{arctg}\frac{x-\frac{1}{x}}{\sqrt{2}} - \frac{1}{2}\cdot\left(-\frac{1}{2\sqrt{2}}\ln\left|\frac{\sqrt{2}+x+\frac{1}{x}}{\sqrt{2}-x-\frac{1}{x}}\right|\right) + C =$$

$$= \frac{\operatorname{arctg}\frac{x-\frac{1}{x}}{\sqrt{2}}}{2\sqrt{2}} + \frac{\ln\left|\frac{\sqrt{2}+x+\frac{1}{x}}{\sqrt{2}-x-\frac{1}{x}}\right|}{4\sqrt{2}} + C$$

5) $\int \frac{x^3+x-1}{(x^2+2)^2}\,dx = \left\{ x^3+x-1 = x(x^2+2)-x-1 \right\} = \int \left(\frac{x}{x^2+2} - \frac{x+1}{(x^2+2)^2} \right) dx =$

$= \int \frac{x}{x^2+2}\,dx - \int \frac{x+1}{(x^2+2)^2}\,dx = \frac{1}{2}\int \frac{d(x^2+2)}{x^2+2} - \int \frac{x+1}{(x^2+2)^2}\,dx = \frac{1}{2}\ln(x^2+2) - \int \frac{x+1}{(x^2+2)^2}\,dx$

$\int \frac{x+1}{(x^2+2)^2}\,dx = \int \frac{x}{(x^2+2)^2}\,dx + \int \frac{dx}{(x^2+2)^2} = \frac{1}{2}\int \frac{d(x^2+2)}{(x^2+2)^2} + \int \frac{dx}{(x^2+2)^2} =$

$= -\frac{1}{2(x^2+2)} + \int \frac{dx}{(x^2+2)^2} = \left\{ \begin{array}{ll} u = \frac{1}{x} & du = -\frac{1}{x^2}\,dx \\ dv = \frac{x}{(x^2+2)^2}\,dx & v = -\frac{1}{2(x^2+2)} \end{array} \right\} =$

$= -\frac{1}{2(x^2+2)} - \frac{1}{2x(x^2+2)} - \int \frac{dx}{2x^2(x^2+2)} = -\frac{1}{2}\left(\frac{1}{x^2+2} + \frac{1}{x(x^2+2)} + \int \frac{dx}{x^2(x^2+2)} \right)$ =

$\frac{A}{x^2} + \frac{Bx+C}{x^2+2} = \frac{Ax^2+2A+Bx^3+Cx^2}{x^2(x^2+2)}$

$\left\{ \begin{array}{l} B=0 \\ A+C=0 \\ 2A=1 \end{array} \right. \quad \begin{array}{l} A=\frac{1}{2} \\ B=0 \\ C=-\frac{1}{2} \end{array}$

$\frac{1}{x^2(x^2+2)} = \frac{1}{2x^2} - \frac{1}{2(x^2+2)}$

$= -\frac{1}{2}\left(\frac{1}{x^2+2} + \frac{1}{x(x^2+2)} + \frac{1}{2}\int \frac{dx}{x^2} - \frac{1}{2}\int \frac{dx}{x^2+2} \right) =$

$= -\frac{1}{2}\left(\frac{1}{x^2+2} + \frac{1}{x(x^2+2)} - \frac{1}{2x} - \frac{1}{2\sqrt{2}} \operatorname{arctg} \frac{x}{\sqrt{2}} \right) + C$

$= \frac{1}{2}\ln(x^2+2) + \frac{1}{2}\left(\frac{1}{x^2+2} + \frac{1}{x(x^2+2)} - \frac{1}{2x} - \frac{1}{2\sqrt{2}} \operatorname{arctg} \frac{x}{\sqrt{2}} \right) + C$